BEFORE YOU READ

## Shapes with their own names

Some matrices are named for where their zeros sit; others for an equation they satisfy. Each name carries a guarantee that simplifies computation.

- L0 **Special matrix shapes**
  - L1 **Named for where the zeros sit**
    - L2 **Diagonal** nonzero entries only on the main diagonal; products, powers and inverses act on the diagonal alone
    - L2 **Identity** diagonal with every diagonal entry 1 $AI = IA = A$
    - L2 **Triangular** zeros above or below the diagonal; determinant and eigenvalues read from the diagonal
  - L1 **Named for an equation they satisfy**
    - L2 **Symmetric** real eigenvalues, orthogonal eigenvectors $A = A^T$
    - L2 **Orthogonal** orthonormal columns; preserves lengths and angles $Q^TQ = I$
  - L1 **More types**
    - L2 **Skew-symmetric**
    - L2 **Nilpotent**
    - L2 **Idempotent**
    - L2 **Permutation**

TAKEAWAY

## The name tells you the shortcut

Spot the shape first: diagonal and triangular matrices give determinants and eigenvalues from the diagonal; symmetric and orthogonal ones bring real eigenvalues and preserved lengths. The full list is on the matrix types page.